Assignment 1

Your Name Here

A microbe called *Cinderspore* is spreading between star systems and dimming stars as it goes. One research vessel, the *Meridian*, was sent out with a small crew to study it and find a way to stop it before it reaches Earth.

The *Meridian*'s crew is busy keeping the ship running and the mission alive while you are analyzing the data that is transmitted by Frank. A few things to know about Frank:

- **Frank is the *Meridian*'s engineer and communications officer.**
- **He gets excited easily and sometimes goes on tangents.**
- **He is good at his job and he trusts you to be good at yours.**
- **Frank is not human.**

Mission Log 001 — from Frank

> Hi hi hi! Okay so. The *Meridian* has been out here a while now and you are the new analyst at Mission Control! I'm Frank and I love data. *Cinderspore* is still spreading and everyone up here is tired and the hull keeps making a noise I don't love, so Mission Control wants a look at how we're all doing. I pulled data from the ship's logs. Please make sense of it.

Variable	Type	Description
`crew_role`	Categorical	Pilot, Engineer, Medic, or Analyst
`ship_module`	Categorical	Where the reading was logged: Engineering, Med Bay, Science Deck, Cargo Bay
`mission_phase`	Categorical	Transit, Orbit, or Planetside
`crew_sleep_hours`	Continuous	Hours slept in the last rest cycle
`hull_temp_f`	Continuous	Hull temperature (°F)
`o2_reserve_pct`	Continuous	Oxygen reserve remaining (%)

Question 0: Data Import

0a. Call in any packages that are needed for this assignment.

0b. Import this week's data here.

Question 1: Describing the Hull Temperature

1a. Find the descriptive statistics for the hull temperature (`hull_temp_f`).

1b. Find the descriptive statistics for the hull temperature (`hull_temp_f`) split by mission phase (`mission_phase`).

1c. Construct the histogram for hull temperature (`hull_temp_f`).

1d. Construct boxplots for each mission phase's (`mission_phase`) hull temperature (`hull_temp_f`).

Question 2: Describing Crew Sleep Hours

2a. Find the descriptive statistics for the number of hours slept in the last rest cycle (`crew_sleep_hours`).

2b. Find the descriptive statistics for the number of hours slept in the last rest cycle (`crew_sleep_hours`) split by crew role (`crew_role`).

2c. Construct the dot plot for number of hours slept (`crew_sleep_hours`). Color the dots by crew role (`crew_role`).

Question 3: Describing Mission Phase

3a. Find the frequency table for the mission phase (`mission_phase`).

3b. Find the contingency table for mission phase (`mission_phase`; use as row) by crew role (`crew_role`; use as column).

3c. Construct a bar graph for mission phase (`mission_phase`).